

A ratio of truncated logarithmic series

the exponents $n + 1$ and n cancel against a telescoping sum

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{\left(1 + x + \frac{x^2}{2} + \cdots + \frac{x^{n-1}}{n-1}\right)^{n+1}}{\left(1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n}\right)^n} dx = 2.$$

Overview. Write

$$S_m(x) = 1 + \sum_{k=1}^m \frac{x^k}{k},$$

so the integrand is S_{n-1}^{n+1}/S_n^n . Two facts settle the problem, and neither requires any asymptotic analysis.

First, the integrand is S_{n-1} times $(S_{n-1}/S_n)^n$, and the second factor is close to 1 because the two polynomials differ by the single term x^n/n . Quantitatively, the defect is bounded by x^n uniformly, which integrates to $\frac{1}{n+1}$.

Second, $\int_0^1 S_{n-1}$ telescopes exactly:

$$\int_0^1 S_{n-1}(x) dx = 2 - \frac{1}{n}.$$

Together these squeeze the integral between $2 - \frac{1}{n} - \frac{1}{n+1}$ and $2 - \frac{1}{n}$, so the limit is 2.

1. Notation and the shape of the integrand

Set

$$S_m(x) = 1 + \sum_{k=1}^m \frac{x^k}{k}, \quad m \geq 0, \quad S_0 \equiv 1.$$

For $x \in [0, 1]$ all terms are non-negative, so $S_m(x) \geq 1 > 0$ and S_m is increasing in m . In particular the quotient

$$r_n(x) := \frac{S_{n-1}(x)}{S_n(x)} \in (0, 1] \tag{1}$$

is well defined, with $r_n(0) = 1$.

Splitting off one factor of S_{n-1} ,

$$I_n := \int_0^1 \frac{S_{n-1}^{n+1}}{S_n^n} dx = \int_0^1 S_{n-1}(x) r_n(x)^n dx. \tag{2}$$

Since $0 < r_n \leq 1$, we get immediately

$$0 < I_n \leq \int_0^1 S_{n-1}(x) dx. \tag{3}$$

Everything now depends on how far r_n^n can fall below 1.

2. The defect is bounded by x^n

The two polynomials differ by exactly one monomial:

$$S_n(x) - S_{n-1}(x) = \frac{x^n}{n}. \tag{4}$$

Lemma. For all $x \in [0, 1]$ and all $n \geq 1$,

$$\boxed{0 \leq S_{n-1}(x)(1 - r_n(x)^n) \leq x^n.} \quad (5)$$

Proof. Write $r = r_n(x) \in (0, 1]$. The finite geometric identity

$$1 - r^n = (1 - r) \sum_{j=0}^{n-1} r^j$$

gives

$$S_{n-1}(1 - r^n) = S_{n-1}(1 - r) \sum_{j=0}^{n-1} r^j.$$

By (1) and (4),

$$S_{n-1}(1 - r) = S_{n-1} \cdot \frac{S_n - S_{n-1}}{S_n} = \frac{x^n}{n} \cdot \frac{S_{n-1}}{S_n} = \frac{x^n}{n} r.$$

Hence

$$S_{n-1}(1 - r^n) = \frac{x^n}{n} r \sum_{j=0}^{n-1} r^j \leq \frac{x^n}{n} \cdot 1 \cdot n = x^n,$$

because every one of the n terms r^j and the prefactor r lie in $(0, 1]$. Non-negativity is clear since $r \leq 1$. $\square$

The lemma is sharp in order: at $x = 1$ one has $S_{n-1}(1) = 1 + H_{n-1} \approx \log n$ and $1 - r_n(1)^n \approx \frac{1}{n} \cdot \frac{n}{1+H_n} \approx \frac{1}{\log n}$, so the left side is of order 1, matching $x^n = 1$.

Integrating (5) over $[0, 1]$,

$$0 \leq \int_0^1 S_{n-1} dx - I_n \leq \int_0^1 x^n dx = \frac{1}{n+1}, \quad (6)$$

that is,

$$\int_0^1 S_{n-1}(x) dx - \frac{1}{n+1} \leq I_n \leq \int_0^1 S_{n-1}(x) dx. \quad (7)$$

3. The telescoping evaluation

The remaining integral is exact:

$$\begin{aligned} \int_0^1 S_{n-1}(x) dx &= 1 + \sum_{k=1}^{n-1} \frac{1}{k} \int_0^1 x^k dx = 1 + \sum_{k=1}^{n-1} \frac{1}{k(k+1)} \\ &= 1 + \sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 + \left(1 - \frac{1}{n} \right) = 2 - \frac{1}{n}. \end{aligned} \quad (8)$$

The harmonic-looking coefficients $\frac{1}{k}$ and the integration of x^k conspire into the telescoping sum $\sum \frac{1}{k(k+1)}$; this is the only place where the specific coefficients of the series matter, and it is what produces the answer $2 = 1 + 1$.

Combining (7) and (8),

$$\boxed{2 - \frac{1}{n} - \frac{1}{n+1} \leq I_n \leq 2 - \frac{1}{n},} \quad (9)$$

so $I_n \rightarrow 2$. Moreover (9) gives the rate:

$$I_n = 2 - \frac{1}{n} + O\left(\frac{1}{n}\right), \quad \text{more precisely} \quad \left| I_n - 2 + \frac{1}{n} \right| \leq \frac{1}{n+1}.$$

n	lower bound $2 - \frac{1}{n} - \frac{1}{n+1}$	upper bound $2 - \frac{1}{n}$	width
5	1.6333	1.8000	0.1667
20	1.9024	1.9500	0.0476
100	1.9801	1.9900	0.0099
1000	1.99800	1.99900	0.00100

4. What the limit “ought” to be, and why the bound is needed

There is a heuristic that gets the answer instantly and hides the only real difficulty. For fixed $x \in [0, 1)$,

$$S_m(x) \longrightarrow 1 + \sum_{k=1}^{\infty} \frac{x^k}{k} = 1 - \log(1 - x),$$

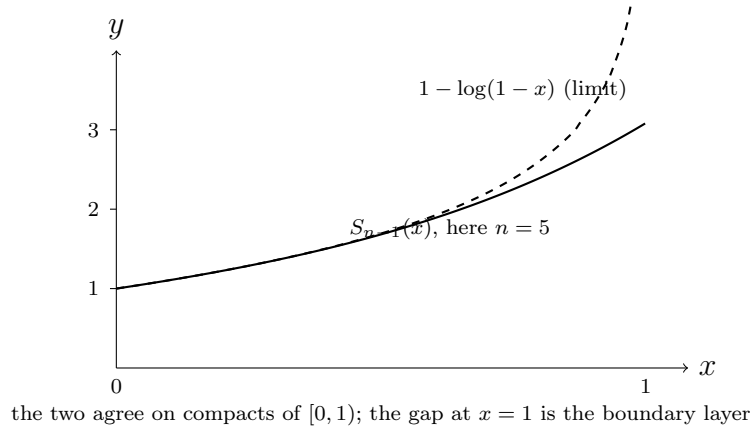
and

$$r_n(x)^n = \left(1 - \frac{x^n/n}{S_n(x)}\right)^n \longrightarrow 1,$$

because $x^n \rightarrow 0$ geometrically. So the integrand converges pointwise on $[0, 1)$ to $1 - \log(1 - x)$, and

$$\int_0^1 (1 - \log(1 - x)) \, dx = 1 + 1 = 2. \quad (10)$$

The heuristic gives the right number, but the convergence is not uniform, and the limit function is unbounded at $x = 1$ while every $\frac{S_n^{n+1}}{S_n^n}$ is a polynomial quotient, bounded on $[0, 1]$. What happens near $x = 1$ is a genuine competition: $S_{n-1}(1) = 1 + H_{n-1} \rightarrow \infty$ like $\log n$, while $r_n(1)^n$ is bounded away from 1. The elementary bound (5) is precisely what controls that boundary layer, and it does so uniformly, with the crude estimate x^n whose integral is $\frac{1}{n+1}$.



Note also that (8) is the finite- n shadow of (10): the exact value $2 - \frac{1}{n}$ converges to the exact value 2 of the limiting integral, with the discrepancy $\frac{1}{n}$ coming from the truncation of the series, not from the exponents.

5. Robustness of the argument

The proof used three properties of the coefficient sequence $c_k = \frac{1}{k}$.

Used where	Property
Lemma, bound $\frac{x^n}{n} \cdot n \leq x^n$	The last coefficient satisfies $c_n \leq \frac{1}{n}$. Any $c_n = O(1/n)$ gives the same conclusion with a constant.
Positivity of $r_n \leq 1$	$c_k \geq 0$, so S_m is increasing in m and bounded below by 1 on $[0, 1]$.
Telescoping (8)	$\int_0^1 x^k dx = \frac{1}{k+1}$ combines with $c_k = \frac{1}{k}$ into $\frac{1}{k(k+1)}$. This is where the value 2 is produced.

Consequently, for any non-negative c_k with $c_n \leq \frac{C}{n}$ and $T_m(x) = 1 + \sum_{k \leq m} c_k x^k$,

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{T_{n-1}^{n+1}}{T_n^n} dx = 1 + \sum_{k=1}^{\infty} \frac{c_k}{k+1},$$

provided the right side converges, by the identical argument. For $c_k = \frac{1}{k}$ the sum is $\sum \frac{1}{k(k+1)} = 1$ and the limit is 2; for $c_k = \frac{1}{k^2}$ it is $1 + \sum \frac{1}{k^2(k+1)} = 1 + \left(\frac{\pi^2}{6} - 1\right) = \frac{\pi^2}{6}$.

Finally, the mismatched exponents $n+1$ and n are a deliberate distraction. Had they been equal, the integrand would be $(S_{n-1}/S_n)^n \rightarrow 1$ and the answer would be 1; the extra factor of S_{n-1} is the entire content, and once (2) is written down the problem is about a single polynomial, not a quotient.